

YAO(JUSTIN) LIU

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EDUCATION

University of California, Berkeley

Starting Aug. 2026

Ph.D. Electrical Engineering & Computer Sciences (EECS)

University of Southern California

Aug. 2021 – May 2025

B.S. Electrical and Computer Engineering (Honors)

GPA: 3.86

B.S. Applied & Computational Mathematics

Research Interests: Embodied Agents, Online Decision-Making, Multi-Agent Systems, Cyber-Physical Systems

Relevant Coursework: Probability Theory, Statistics, Distributed Systems & IoT, Real Analysis, PDE, Numerical Methods, Machine Learning, Reinforcement Learning (Coursera), Linear Systems, Algorithms, OOP, Embedded Systems, Digital Circuits

EXPERIENCE

Design Automation of Intelligent Systems Lab, Northwestern University

July 2024 — Present

Research Assistant, Prof. Qi Zhu's Group

- **Multi-Agent Systems via Diffusion Predictive Modeling and Selective Communications:**

Co-developed a conditional diffusion model that infers global state from local views and limited communication for a team of heterogeneous robots; designed a MARL framework enabling heterogeneous robots to coordinate resource exploration, extraction, and transport in NASA's ISRU challenge; co-designed a closed-loop system where diffusion-based predictions guide MARL policies, reducing inter-agent communication by leveraging diffusion inference uncertainty.

- **Safe Embodied LLM-Agent via Temporal Logic:**

Developed SENTINEL, a multi-level safety framework for embodied LLM agents that integrates temporal logic safety constraints into task planning, with semantic, plan, and trajectory-level verification; Enabled LLMs for collision-aware navigation planning in physics-informed environments; Implemented computation-tree-based checking in VirtualHome and AI2-THOR to detect unsafe states, ordering violations, and timed hazards with interpretable counterexamples.

Autonomous Networks Research Group, USC

Summer 2024 — Present

Research Assistant, Prof. Bhaskar Krishnamachari's Group

- **Fairness in Cooperative Multi-Agent Systems:**

Designed a multi-agent RL framework to learn Eisenberg–Gale equilibrium task allocation for fair and efficient agent–task service under partial observability; Integrated attention-based GNN message passing to enable relational reasoning over local agent–entity interactions; Developed a stochastic online algorithm that dynamically assigns agents to sequentially discovered goal sets based on the Eisenberg–Gale convex program.

Computational Optimization Laboratory, Stanford University

Summer 2024 — Present

Research Assistant, Prof. Yinyu Ye's Group

- **Arrow-Debreu Competitive Equilibrium: Online Stochastic Optimization:**

Developed action-history-dependent online optimization algorithms via certainty-equivalent & sample average approximation methods to sequentially compute equilibrium prices and allocations under incomplete agent information; Established regret $O(\log n)$ and constraint violation $O(1)$, demonstrating enhanced performance over static algorithms.

Energy Efficient Secure Sustainable Computing Group, USC

Aug. 2023 — Present

Research Assistant, Prof. Peter Beerel's Group

- **Conditional Diffusion Framework for Wildfire Origin and Spread Prediction:**

Co-designed a conditional diffusion model integrating ALERTCalifornia video, active-fire maps, and geospatial layers with a cross-attention UNet to infer ignition and spread; Collaborated on formulating a unified bird's-eye-view representation of fire scenes via camera-pose-based projection; Designed a real-time Unreal Engine rendering system for wildfire spread.

- **Distributed Inference & Adversarial Detection:**

Co-developed a dynamic partial-redundancy scheme for secure distributed ML inference robust against Byzantine faults; Derived closed-form bounds on the inferences needed for 99% attack detection confidence under low redundancy (2-5%) and proved a negligible detection gap between independent and centralized RNG schedulers, validated on GPT-2.

Stanford University

Nov. 2025 – Jun. 2026

- **AnyMug: Pose-Agnostic Robotic Functional Grasping:**

Co-developed a closed-loop PPO framework that jointly canonicalizes depth observations and 7D end-effector actions in a mug-centric frame; incorporated handle-aware rewards, pose curriculum learning, and domain randomization for sim-to-real; achieved 93%+ success in simulation and 80% zero-shot success on unseen mugs using a Franka Panda.

Radio Lab, Yale University

Sep. 2025 — Present

Research Assistant, Prof. Dionysis Kalogerias' Group

- **Zeroth-Order Robust Multi-Agent Policy Learning:**

Formulated a min-max adversarial multi-agent reinforcement learning algorithm that enables decentralized agents to

remain robust against targeted observation attacks; Integrated zeroth-order gradient estimators to optimize policies for non-differentiable, non-convex objectives under worst-case observation perturbations.

ACADEMIC SERVICE

Conference Reviewer, IEEE Intelligent Vehicles Symposium (IV)	2026
Conference Reviewer, Conference on Robot Learning (CoRL)	2026

PUBLICATIONS (*Indicates equal contribution; †Indicates alphabetical order)

1. **Yao Liu**, S. Mohanty, E. Ondula, Bhaskar Krishnamachari. *DISPATCH: Decentralized Informed Spatial Planning and Task Assignment for Cooperative Heterogeneous Agents*. Submitted to **IEEE RAL 2026**. <https://arxiv.org/abs/2511.17915>
2. Simon S. Zhan*, **Yao Liu***, Philip Wang*, Z. Wang, Q. Wang, Z. Ruan, X. Shi, X. Cao, Frank Yang, K. Wang, Huajie Shao, Manling Li, Qi Zhu. *SENTINEL: A Multi-Level Formal Framework for Safety Evaluation of LLM-based Embodied Agents*. Under Review. <https://nu-ideas-lab.github.io/Sentinel>
3. Xuan Zhou, U. Muhan, **Yao Liu**, Peter Beerel. *An Efficient Distributed Inference Framework for IoT Systems with Byzantine Fault Detection*. **GLSVLSI 2025**. <https://dl.acm.org/doi/10.1145/3716368.3735230>
4. Yue Hu, G. Datta, K. Beerel, **Yao Liu**, Peter Beerel. *Let's Roll: Synthetic Dataset Analysis for Pedestrian Detection Across Different Shutter Types*. **IEEE SiPS 2024**. <https://ieeexplore.ieee.org/stamp/stamp.jsp?arnumber=10768224>
5. Anthony Goeckner, Weihe Gao, **Yao Liu**, Alessandro Pinto, Qi Zhu. *Quiet Multi-Agent Systems via Predictive World Modeling and Selective Communications*. In submission.
6. Zizhan Tang*, **Yao Liu***, Jiexi Liu. *Predictive Modeling in AUV Navigation: A Perspective from Kalman Filtering*. Submitted to **IEEE IV 2026**.
7. **Yao Liu**, Bryan Shaddy, Assad Oberai, Peter Beerel. *Visualization of Wildfire Spread and Smoke Using Unreal Engine*. In preparation.
8. C. Chen†, **Yao Liu†**, J. Yang†, Z. Yang†, Yinyu Ye. *Adaptive Online Pricing for Arrow–Debreu Markets*. In preparation.
9. Le Qiu, Cole Harrison, Jiankai Sun, **Yao Liu**, Suning Huang, Qianzhong Chen, Yang You, Marco Pavone. *Pose-Agnostic Robotic Functional Grasping via Observation–Action Canonicalization*. arXiv preprint <https://arxiv.org/abs/2606.21148>.
10. Xi Jia, X. Zhang, C. Feng, L. Liu, **Yao Liu**, A. Xia, Yue Ma. *Exploring the Associations between Teacher and Student Depression: Evidence from Rural Primary Schools in China*. Under review at the **International Journal of Education**.

TEACHING EXPERIENCE

Course Producer, EE 202: Linear Circuits, USC	Fall 2023 – Spring 2025
Led lab sections, delivered hands-on guidance, and explained course concepts through office hours & exam review.	
Course Grader, EE 301: Linear Systems; MATH 245: Math of Phys & Eng I, USC	Fall 2024
Graded homework assignments & lab reports.	

PROJECT EXPERIENCES

LMSYS Chatbot Arena Human Preference Predictions	Summer 2024
<i>Kaggle Competition</i>	

- Developed a robust reward modeling system to predict human preferences of LLMs (Gemma-2-9b and Llama-3.1-8b) using advanced Parameter-Efficient Fine-Tuning techniques; Designed a weighted ensemble inference engine that aggregated model predictions, significantly reducing log-loss against the competition baseline.

Predictive Modeling for Safe AUV Navigation	Spring 2024
<i>International Mathematical Contest in Modeling</i>	

- Developed a safe navigation pipeline for AUVs by combining multi-buoy TDOA multilateration with Kalman filtering to achieve noise-robust localization and trajectory prediction; formulated scenario-specific uncertainty-propagation models (normal navigation vs. propulsion failure) to generate principled, time-varying search regions for recovery operations.

HONORS and GRANTS

Discover Scholar Distinction, USC	Spring 2025
Gold Medalist, Kaggle—Chatbot Arena Human Preference Predictions	Summer 2024
Honorable Mention, Mathematical Contest in Modeling (COMAP)	Spring 2024
Ming Hsieh Institute Undergraduate Scholar, USC (top 5 research scholars in Electrical Engineering)	Spring 2024
Academic Achievement Award, USC	Spring 2024 – Spring 2025
Provost Research Scholarship, USC	Spring 2024
W.V.T. Rusch Engineering Honors Degree, USC	Fall 2023
Richardson CURVE Award, USC	2022 – 2024
Dean's List, USC	Spring 2022 – Spring 2025

PROFESSIONAL SKILLS

Languages: Python (NumPy, CVXPY, TensorFlow, PyTorch), C/C++, Verilog & HDL, MATLAB, HTML

Robotics: OpenCV, OpenAI Gymnasium, Alfred, ROS2 VirtualHome, BEHAVIOR-1K, MPE, MuJoCo, NVIDIA Isaac Sim

Machine Learning: Reinforcement Learning, Diffusion Models, Online Learning, SGD, Adaboost, Transformer

DevOps/Tools: Git, CVXPY, Unity, Unreal Engine, Simulink, Docker, AWS, Firebase, REST APIs, MySQL